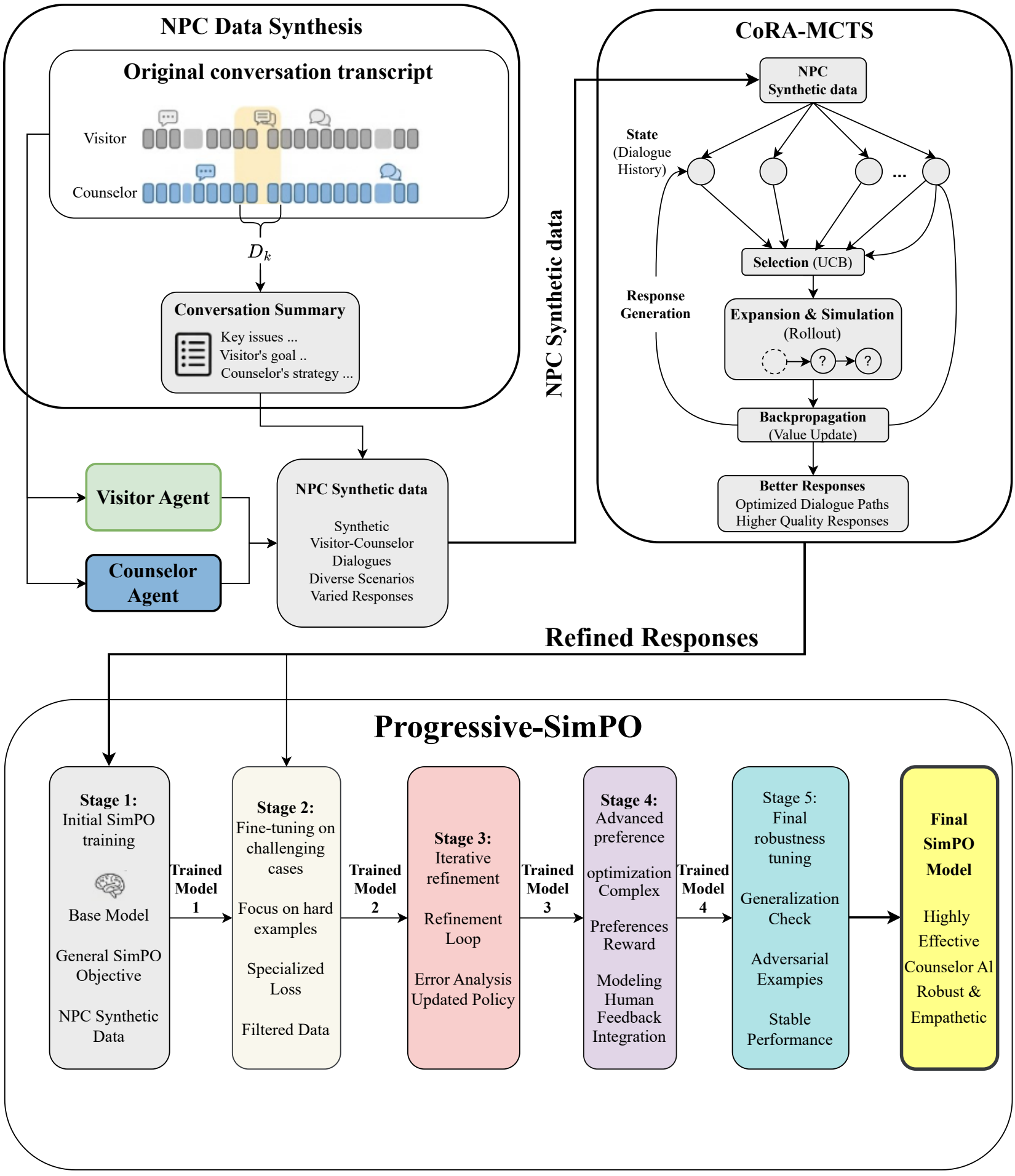
NPC Data Synthesis
Original conversation transcript
Visitor
Counselor
D_k
Conversation Summary
Key issues ...
Visitor's goal ..
Counselor's strategy ...
Visitor Agent
Counselor Agent
NPC Synthetic data
Synthetic
Visitor-Counselor
Dialogues
Diverse Scenarios
Varied Responses
NPC Synthetic data
CoRA-MCTS
NPC
Synthetic data
State
(Dialogue
History)
...
Selection (UCB)
Response
Generation
Expansion & Simulation
(Rollout)
?
?
Backpropagation
(Value Update)
Better Responses
Optimized Dialogue Paths
Higher Quality Responses
Refined Responses
Progressive-SimPO
Stage 1:
Initial SimPO
training
Base Model
General SimPO
Objective
NPC Synthetic
Data
Trained
Model
1
Stage 2:
Fine-tuning on
challenging
cases
Focus on hard
examples
Specialized
Loss
Filtered Data
Trained
Model
2
Stage 3:
Iterative
refinement
Refinement
Loop
Error Analysis
Updated Policy
Trained
Model
3
Stage 4:
Advanced
preference
optimization
Complex
Preferences
Reward
Modeling
Human
Feedback
Integration
Trained
Model
4
Stage 5:
Final
robustness
tuning
Generalization
Check
Adversarial
Exampies
Stable
Performance
Final
SimPO
Model
Highly
Effective
Counselor AI
Robust &
Empathetic